

PRANAV JAIN

University of Southern California

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[Website](#) , [Scholar](#)

RESEARCH INTERESTS

Computer Graphics, Physical Simulations, Geometry Processing, Numerical Analysis of Partial Differential Equations, Discrete Differential Geometry

EDUCATION

University of Southern California (USC) Doctor of Philosophy (PhD), Computer Science	August 2023 - Present <i>Los Angeles, USA</i>
New York University (NYU) Master of Science (MS), Scientific Computing	September 2021 - May 2023 <i>New York, USA</i>
Indraprastha Institute of Information & Technology Delhi (IIITD) Bachelor of Technology with Honors (B-Tech Hons), Computer Science and Applied Mathematics	August 2016 - August 2020 <i>New Delhi, India</i>

PUBLICATIONS

- Pranav Jain**, Navami Kairanda, Peter Yichen Chen, & Oded Stein. (2026). PINNsur: Physics-Informed Neural Networks for PDEs on Curved Surfaces. [\[link\]](#)
- Pranav Jain**, Ziyin Qu, Peter Yichen Chen, and Oded Stein. 2024. Neural Monte Carlo Fluid Simulation. In ACM SIGGRAPH 2024 Conference Papers (SIGGRAPH '24). [\[link\]](#)
- Liam Martin, **Pranav Jain**, Zachary Ferguson, Torkan Gholamalizadeh, Faezeh Moshfeghifar, Kenny Erleben, Daniele Panozzo, Steven Abramowitch, Teseo Schneider. A systematic comparison between FEBio and PolyFEM for biomechanical systems. In Computer Methods and Programs in Biomedicine, 2023. [\[link\]](#)
- Zachary Ferguson, **Pranav Jain**, Denis Zorin, Teseo Schneider, and Daniele Panozzo. High-Order Incremental Potential Contact for Elastodynamic Simulation on Curved Meshes. In ACM SIGGRAPH 2023 Conference Proceedings (SIGGRAPH '23). [\[link\]](#)
- Pranav Jain**, Munawar Hasan, Donghoon Chang. Spy based analysis of selfish mining attack on multi-stage blockchain. In Cryptology ePrint Archive, 2019. [\[link\]](#)

RESEARCH EXPERIENCE

- | | |
|--|---|
| Google Summer of Code [link]
<i>Contributor</i>
Advisor: Prof. Martin Skrodzki, Dr. Andreas Fabri | May 2026 - August 2026
<i>Remote</i> |
|--|---|
- Implemented the algorithm proposed in *Lipschütz et al. (2024). Manifold Meshing of Point Clouds with Guaranteed Smallest Edge Length* for mesh reconstruction from scratch and integrated it into the Computational Geometry Algorithms (CGAL) library
- | | |
|---|---|
| Adobe Research
<i>Research Intern</i>
Advisor: Dr. Timothy Langlois, Dr. Danny Kaufman, Dr. Jui-hsien Wang | May 2025 - August 2025
<i>Seattle, USA</i> |
|---|---|

- Researched on how to make generative-AI models more physics accurate by limiting hallucinations for video generation

University of Southern California

PhD Student

August 2023 - Present

Los Angeles, USA

Advisor: Prof. Oded Stein, Prof. Krishna Garikipati

- Researching on empirically proving convergence rates of Poisson problem using Physics Informed Neural Networks (PINN) with respect to the size of the architecture
- Researched and published on a discretization-free fluid simulation method using combination of Monte-Carlo and neural fields

New York University

Research Assistant

September 2021 - May 2023

New York, USA

Advisor: Prof. Daniele Panozzo, Prof. Denis Zorin

- Researched and compared the simulation abilities of PolyFEM and FEBio in terms of accuracy, time and user inputs
- Introduced a high-order finite element formulation for elastodynamic simulation with contact handling

nTopology

Software Engineer Intern in Geometry Team

June 2022 - August 2022

New York, USA

Advisor: Suraj Musuvathy, Ranjeeth Mahankali

- Designed and implemented a method to efficiently stitch different CAD models to form one unified CAD structure

Freie Universität Berlin [\[link\]](#)

Research Intern

September 2020 - August 2021

(Virtual) Berlin, Germany

Advisor: Prof. Konrad Polthier, Dr. Sunil Kumar Yadav

- Researched on a robust and parameter-tuning free method to denoise a point cloud

Fields Undergraduate Summer Research Programme 2020 [\[link\]](#)

Research Intern

July 2020 - August 2020

(Virtual) Toronto, Canada

Advisor: Prof. Thomas Uchida

- Researched on the problem of inverse mechanism synthesis – given a path, design a mechanism that traces the path

Indraprastha Institute of Information & Technology Delhi [\[link\]](#)

Research Assistant

August 2018 - August 2020

New Delhi, India

Advisor: Prof. Kaushik Kalyanaraman

- Developed DECAT, a software framework for discretizations of the objects and operators of exterior calculus

Indraprastha Institute of Information & Technology Delhi [\[link\]](#)

Undergraduate Thesis

August 2018 - August 2020

New Delhi, India

Advisor: Prof. Donghoon Chang

- Proved the selfish mining attack strategy on the Bitcoin blockchain for multiple mining pools
- Designed and proved an extended attack for multi-stage blockchain

ACADEMIC COMMUNITY WORK

Reviewing

- SIGGRAPH 2026 February 2026
- SIGGRAPH Asia 2025 July 2025
- Pacific Graphics (PG) 2025 July 2025

Volunteering

- Teaching assistant for the Poisson Reconstruction Project at Summer Geometry Initiative (SGI) 2024 July 2024 - August 2024
- Former member of the Conference Coffee team at SIGGRAPH Research Career Development Committee. 2021-2022

TEACHING EXPERIENCE

University of Southern California

- Teaching Assistant: CSCI 532 - Innovation for Defense Applications Spring 2026
- Teaching Assistant: CSCI 103 - Introduction to Programming Fall 2025
- Teaching Assistant: CSCI 580 - 3D Graphics and Rendering Spring 2025
- Teaching Assistant: CSCI 596 - Scientific Computing and Visualization Fall 2024
- Teaching Assistant: CSCI 104 - Data Structures and Object Oriented Design Spring 2024

New York University

- Grader: MATH 263.3 - Applied Partial Differential Equations Spring 2023
- Grader: MATH 252 - Numerical Analysis Spring 2023
- Grader: MATH 263 - Partial Differential Equations Fall 2021